

GenAI Literacy: A Framework for Higher Education Instructors

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Since it became widely accessible in 2022, Generative AI (GenAI) has emerged as one of the most significant technological milestones to impact higher education. While the debate rages on about whether and how much we should allow GenAI into our classrooms, it is undeniable that we are all living in a world with proliferating GenAI, whether we like it or not. We do not yet know the kind of future we are heading toward, nor do we fully understand the costs GenAI will pose to people, culture, politics, or the planet. But we are living in this world despite these unknowns, and so are our students.

How can we prepare our students to live, learn, and succeed in this world? Moreover, how can we prepare them to make nuanced, informed decisions in this world? How can we prepare them to successfully make demands of an AI-driven economy and those who wield power within it? This moment calls for a focus on critical literacy with Generative AI. We must engage our students to think deeply about these tools, the broader sociopolitical dynamics behind them, and the ways students can be empowered to be conscious consumers and/or abstainers of GenAI.

Begin Anywhere: Getting Started with GenAI Literacy

This resource synthesizes multiple frameworks for GenAI literacy into a streamlined and accessible model. The goal of this resource is not to provide a definitive framework, advance a singular perspective, or argue for a timeless or superior approach to GenAI literacy. Generative AI tools are still becoming widely popularized and are being developed and studied at a rapid pace. Many frameworks and perspectives for AI literacy are currently in development, and many frameworks can guide educators as they try to meet the demands of this moment.

In this resource, I attempt to put multiple AI literacy frameworks into conversation in a way that provides a clear invitation for instructors to ***begin anywhere***. Cultivating students' critical AI literacy does not need to be hopelessly complicated, nor do instructors need to become subject matter experts in machine learning to begin teaching their students about AI. Like most critical competencies, students can and should learn AI literacy from a variety of perspectives and approaches.

In what follows, I synthesize many different perspectives on AI literacy to find points of overlap, intersection, and interstitial opportunities. From this synthesis, I preview a framework that offers several entry points for cultivating students' AI literacy, in particular focusing on cultivating literacy with Generative AI.

I emphasize a no-wrong-door approach to getting students to think critically about GenAI. Whether an instructor wants to ban or embrace students' use of GenAI, this framework provides an opportunity to bring students into deeper thinking about living, learning, and working in a world characterized by widespread AI.

Major Themes in Existing Frameworks

AI has been a technological cornerstone for nearly two decades, and attempts to build students' AI literacy skills have been theorized for many years. Many of these frameworks build on existing digital and information literacy frameworks, which emphasize that information (including digitally-mediated information) is contextual, bound up in power relations, requires critical discernment, and, especially in the case of digital literacy, requires some technological proficiencies to access and make use of information ecosystems (ACRL, 2016; ALA, 2011).

Progressively Complex Thinking

Many AI literacy frameworks use Bloom's taxonomy as an organizing heuristic, emphasizing that AI literacy engages different and progressively more complex cognitive dimensions. For example, Ng et al. (2023) specify "Cognitive domains" as a functional area of AI literacy in which students are able to know and understand AI and its societal implications, use and apply AI for different purposes and circumstances, and evaluate and create AI to solve problems. Similarly, the University of Adelaide (n.d.) explicitly aligns their AI literacy framework with Bloom's taxonomy, arguing that AI literacy entails being able to recognize and understand AI, use and apply AI, and evaluate and critique AI. Hervieux and Wheatley (2024) also align their framework to Bloom's taxonomy and the ACRL information literacy framework by scaffolding AI literacy as knowing principles of AI, understanding different types of AI, experimenting with AI tools (apply), reviewing outputs of AI tools (analyze), evaluating the impact of AI on society, and engaging with AI discourse.

AI Ethics

In addition to progressively complex cognitive dimensions, many frameworks emphasize ethics as a distinct component of AI literacy. In both frameworks referenced above, authors set apart AI ethics as a distinct pillar of AI literacy that not only includes ethical use of AI tools, but also the ability to pay attention to and formulate a balanced view of the ethical issues engendered by AI. Other frameworks position critical thinking (Biagini et al., 2024) and responsible use (Chiu et al. 2024) under the banner of AI ethics. Meanwhile, Cardon et al. (2023) use the more general frame of "accountability" to describe responsible and ethical use of AI.

Operational Proficiency

Most frameworks also emphasize that AI literacy requires some level of technical or operational proficiency - the ability to interact with AI tools effectively to achieve desired

output that is relevant to the purpose or context. In other words, students must develop technical skills for the practical application and use of AI tools. For example, some frameworks emphasize that students need to know how to collaborate with AI technologies (Chiu et al, 2024), while others emphasize knowing how to work with AI tools to solve problems (Biagini et al., 2024) and exercising awareness when selecting and aligning an AI tool to a task (Cardon et al., 2023).

Agency

Finally, many frameworks emphasize some dimension of agency and authenticity, the idea that students must remain vigilant, critical, and self-regulating when interacting with AI. Though many frameworks feature an agentic dimension, they use different terms to do so. For example, Cardon et al. (2023) specify that AI literacy entails authenticity (prioritizing genuine communication), accountability (taking responsibility for AI-influenced content), and agency (retaining control and practicing self-regulation). Chiu et al. (2024) emphasize self-reflection as a pillar for AI literacy, meaning students can reflect on their current levels of understanding of AI and engage in future learning. Hervieux and Wheatley (2024) position agency as the ability to engage in AI discourse and stay involved in emergent conversations and developments in the AI field.

Critical Awareness

A necessary component of critical literacy is the ability to set information and texts within the broader social, political, and cultural contexts from which they emerge. Extending this to critical AI literacy, students should be able to recognize the forces that are acting upon the AI industry and, in turn, the ways the AI industry shapes and is shaped by society. Hervieux and Wheatley (2024) emphasize that students must be able to comment on and critique AI bias and its impact on the environment. Bali (2023) argues that critical AI literacy must also incorporate tenets of critical pedagogy, focusing on power, justice, and inequities that are reflected in and reproduced by AI.

Presenting a Synthesized Framework

Drawing on the major themes above, critical GenAI literacy involves the following key skills:

- **Awareness:** Students develop an understanding of what GenAI is, how it works, and what it means for humans and the environment. Awareness also includes a broader critical awareness of the human actors behind AI and the social, economic, and political forces that act upon the AI industry and, as a result, shape the speed and scope of development of GenAI tools.
- **Application:** Students know how to use GenAI, how to experiment with different AI tools and techniques, and how to align the tool and interaction to their larger task or purpose. On the input side of the interaction, application entails knowing techniques for prompting and iterating with a GenAI tool. On the output side,

application includes developing the habit of critical discernment when encountering and using GenAI-generated information.

- **Accountability:** Students learn how to take responsibility for their use of GenAI, how to properly attribute or acknowledge their use of GenAI, how to use GenAI in a way that does not violate principles of integrity or authenticity, and are aware of the ethical implications and unresolved questions related to AI.
- **Agency:** Students retain and protect their own self-efficacy, authenticity, and identity when using GenAI tools. Agency includes being able to practice metacognition and self-regulation when using GenAI, which in turn requires students to perceive when they are offloading *too much* cognitive work to the tool. Students with GenAI agency can also develop an informed, balanced personal stance on AI that reflects their unique values, needs, and interests.

All of these skills are supported by scaffolded teaching and learning activities that can align to Bloom's taxonomy and position students to use and think about GenAI in progressively complex ways. From learning basic AI terminology, to experimenting and critiquing GenAI output, to co-constructing projects with the support of a GenAI tool, learning activities can be scaffolded to lead students through progressively complex thinking in any of these skill areas.

A Compass for Navigating GenAI Literacy

GenAI literacy skills and the scaffolded activities that support them are not necessarily discrete categories developed one at a time. In practice, these skill areas are overlapping, mutually reinforcing, and mutually constitutive.

For example, an activity that asks students to prompt a GenAI chatbot and evaluate its output (application) will also benefit from a primer on the limitations and biases of text prediction models (awareness), discussing the extent to which a tool captures students' original intent (agency), and/or reflection on how students might act responsibly with AI during the course (accountability).

Rather than see these skills as discrete categories, it can be beneficial to see them as a kind of compass. Similar to how one might use a compass to navigate a new terrain, students and instructors can use any of these skills as an entry point for navigating GenAI literacy (see Figure 1 below). Just as a compass can help a navigator find their location relative to cardinal directions, this approach to GenAI literacy can help instructors situate their GenAI activities within a broader set of GenAI literacy themes and considerations.

Figure 1: GenAI Literacy Compass
(see below for detailed image description)



Figure 1 Image Description

A circle with 4 points at the top, bottom, left side, and right side, as well as a smaller circle at the center which reads "GenAI Literacy." Within the larger circle, each quadrant has a gradient color that blends with the colors of each adjacent quadrant. The 4 points of the circle represent four AI literacy skill areas: Awareness (what GenAI is, how it works, how it came to be), Application (how to use GenAI tools in various tasks and use cases), Accountability (acting responsibly with AI and AI-influenced content/interactions), and Agency (retaining sense of control, authenticity, and purpose). Between each of these points are example topics that sit in between these skill areas. Between Awareness (top) and Application (left), topics include history and types of AI, model bias and limitations, and principles of prompting. Between Application (left) and Accountability

(bottom), topics include critical evaluation of output and attribution and academic integrity. Between Accountability (bottom) and Agency (right), the topic includes making ethical choices. Between Agency (right) and Awareness (top), topics include consumer choice and advocacy and alignment to personal values. The larger circle is outlined with a dotted line and the words “Scaffolded Opportunities,” indicating that scaffolded opportunities encircle the framework and can apply to all four skill areas and the topics in between.

An Invitation to Begin Anywhere

In this environment of proliferating Generative AI, it can be daunting to consider how to talk with students about AI, let alone integrate these tools into teaching and learning. This framework offers an invitation to begin anywhere, recognizing that many different topics, approaches, exercises, and activities can help cultivate students’ GenAI literacy.

Instructors might believe that they need to be experts in AI before they can begin engaging students with and/or about these technologies. But GenAI literacy can and should be cultivated from a variety of perspectives and disciplinary points of view. Students should not learn about AI solely from AI experts, though such expertise would certainly help them understand how these models work. Crucially, students also need to learn about AI literacy from writers, artists, historians, social scientists, and other disciplinary mentors who can help guide their thinking and practice with AI, and also without it.

Instructors might also believe that they must draw a line in the sand with GenAI, either wholeheartedly embracing it or banning it completely. Neither of these two extremes will help students develop deeper thinking or balanced points of view for living and learning in a world with AI. The dichotomy between ban or embrace is a false choice. Instructors can teach students *about* AI without necessarily allowing wholesale use of these tools. Similarly, instructors can scaffold students into particular uses of these tools while also teaching about their limitations and ethical challenges.

Whether an instructor decides to teach *about* AI or teach *with* AI is ultimately a personal decision. As Bali (2023) thoughtfully articulates, when it comes to teaching about, or teaching with, AI, “there isn’t one right way, there is the way that works for you within the parameters of your situation.” There are many ways to teach students about and with GenAI that can cultivate their critical literacy, instructors need only to begin.

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